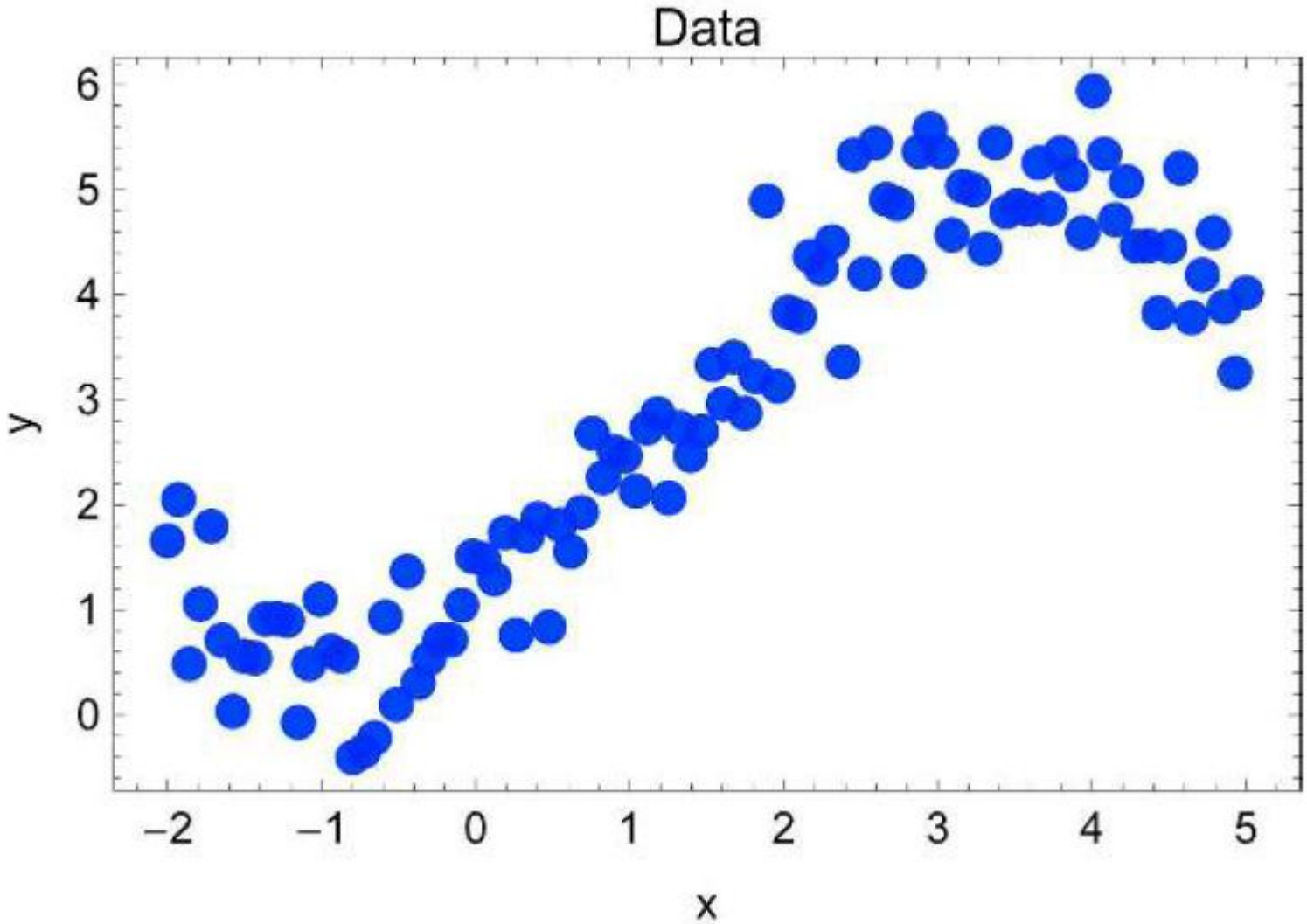
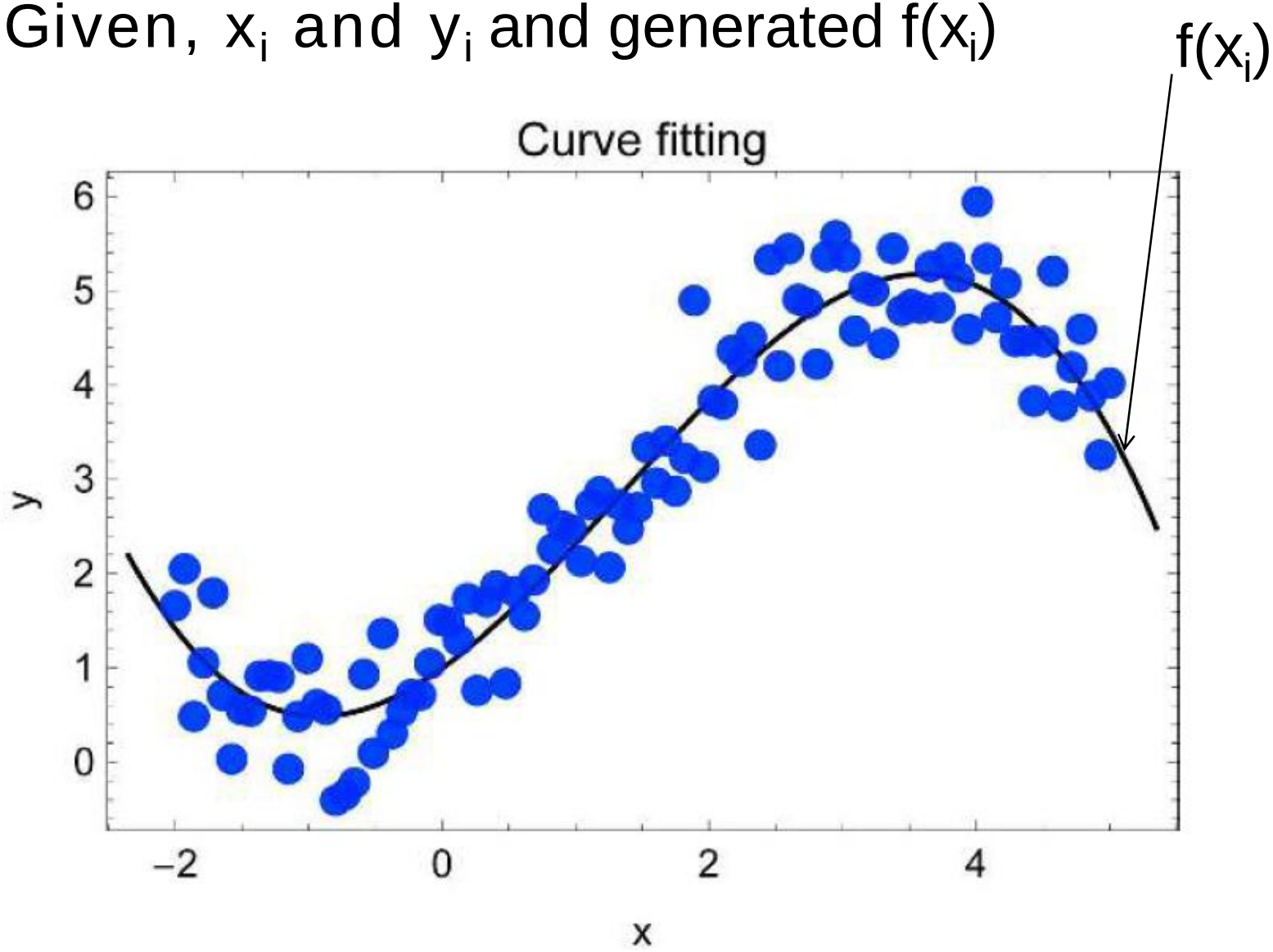
Adaptive system and cost function

R Srinivasan
Faculty
Department of Information Technology
SSN College of engineering
OMR, Kalavakkam-603 110

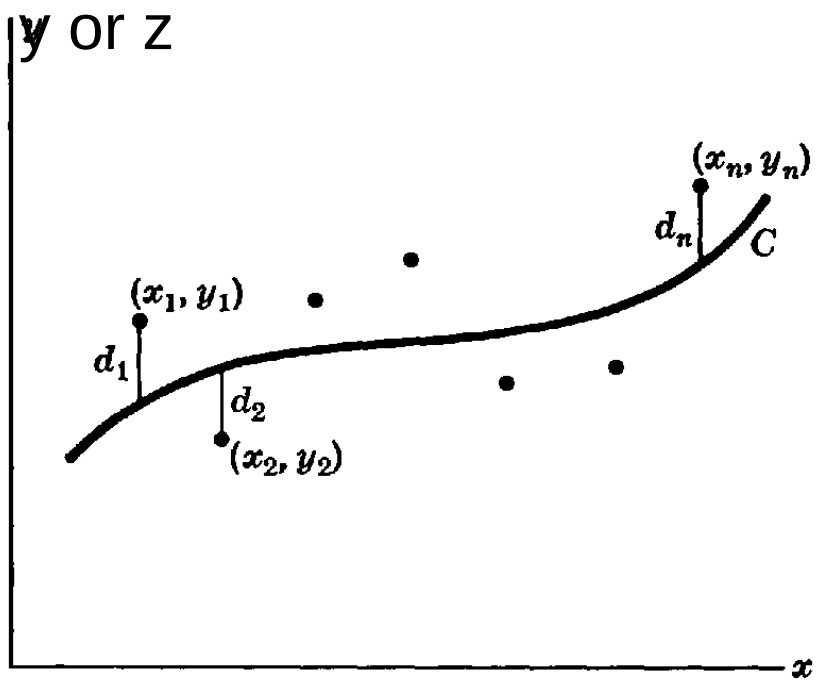
- Given, x_i and y_i
- K number of data points



- Given, x_i and y_i and generated $f(x_i)$



Deviation - error

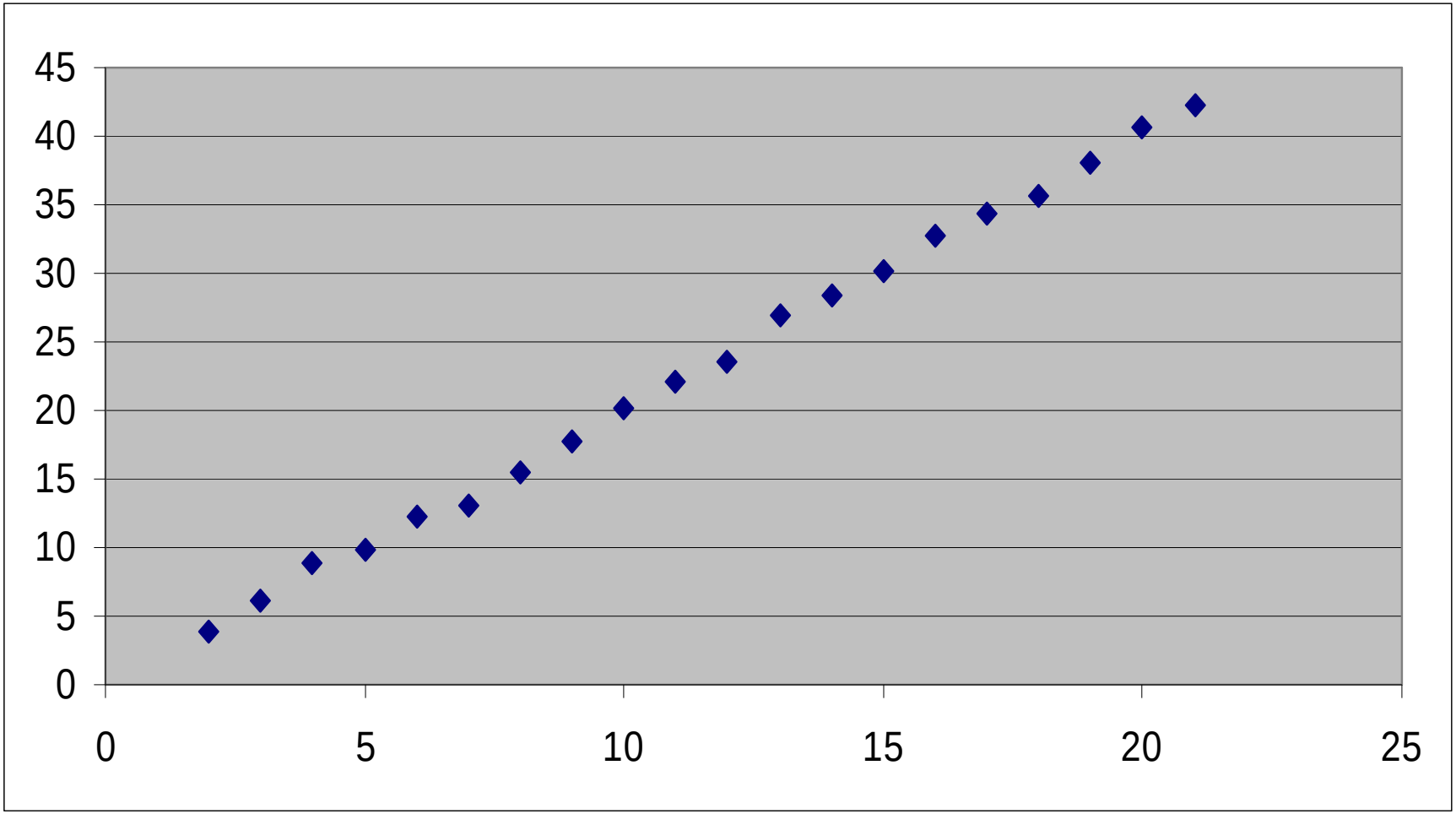


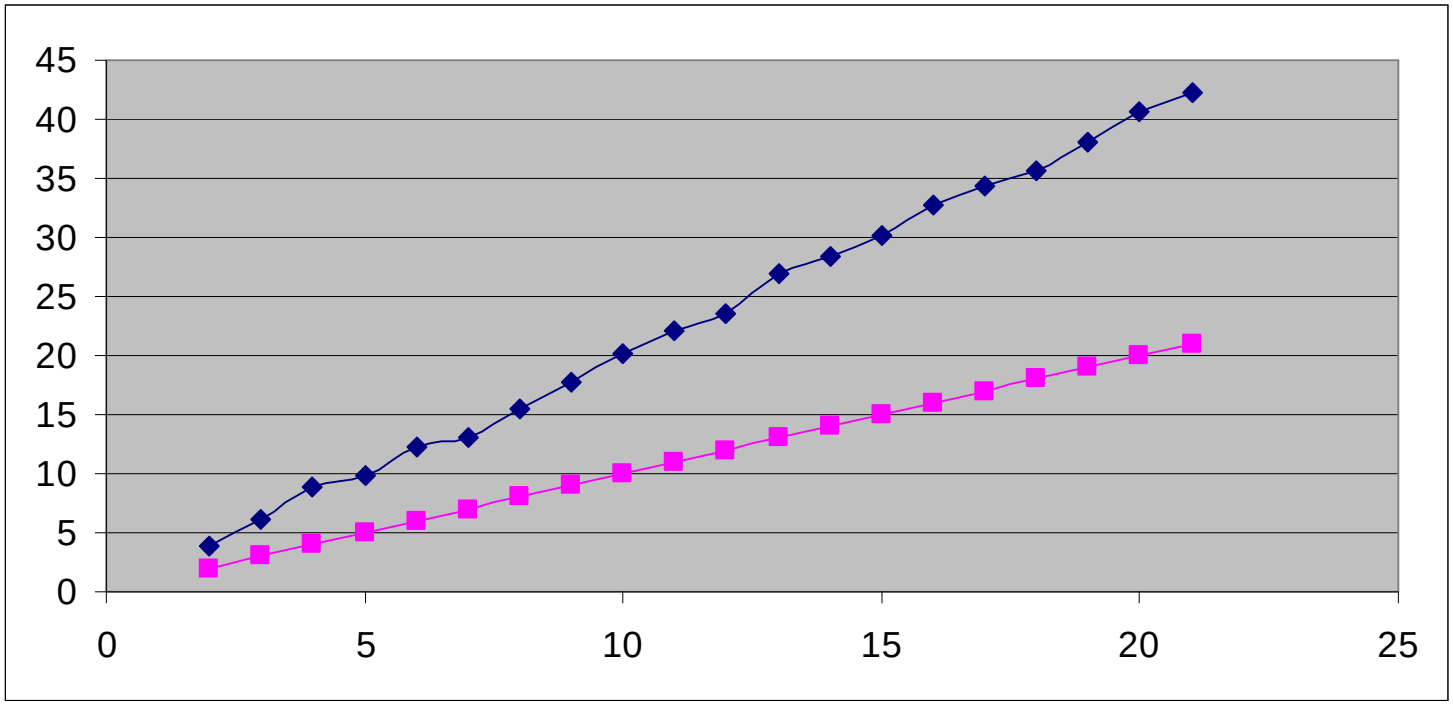
- Given, x_i and y_i
- K number of data points
- Error = $y_i - f(x_i)$
- How many errors?
- K errors

Sum of Squared errors

$$d_1^2 + d_2^2 + \cdots + d_k^2 = \text{a minimum}$$

$$\sum_{i=1}^{K} (y_i - f(x_i))^2 \longrightarrow \text{minimize}$$

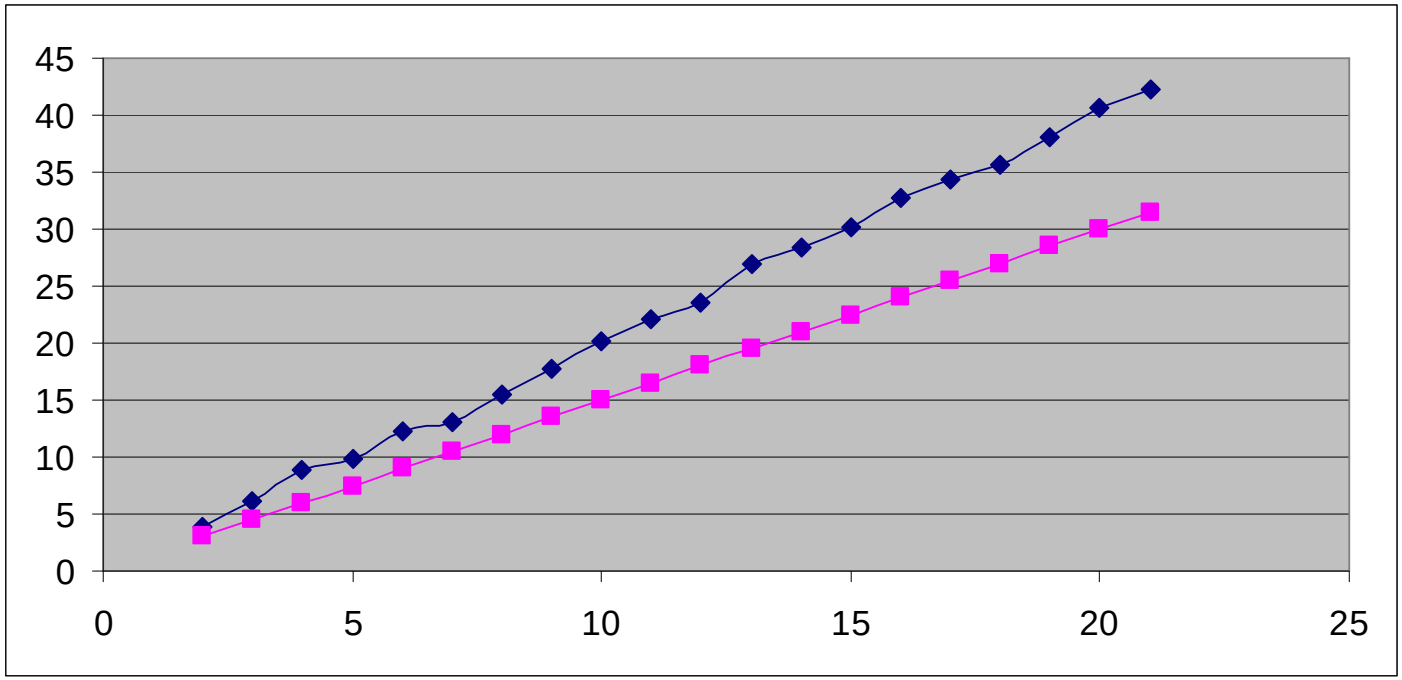

45
40
35
30
25
20
15
10
5
0
0
5
10
15
20
25



Fit with m=1, and c = 0

$$y = m.x + c$$

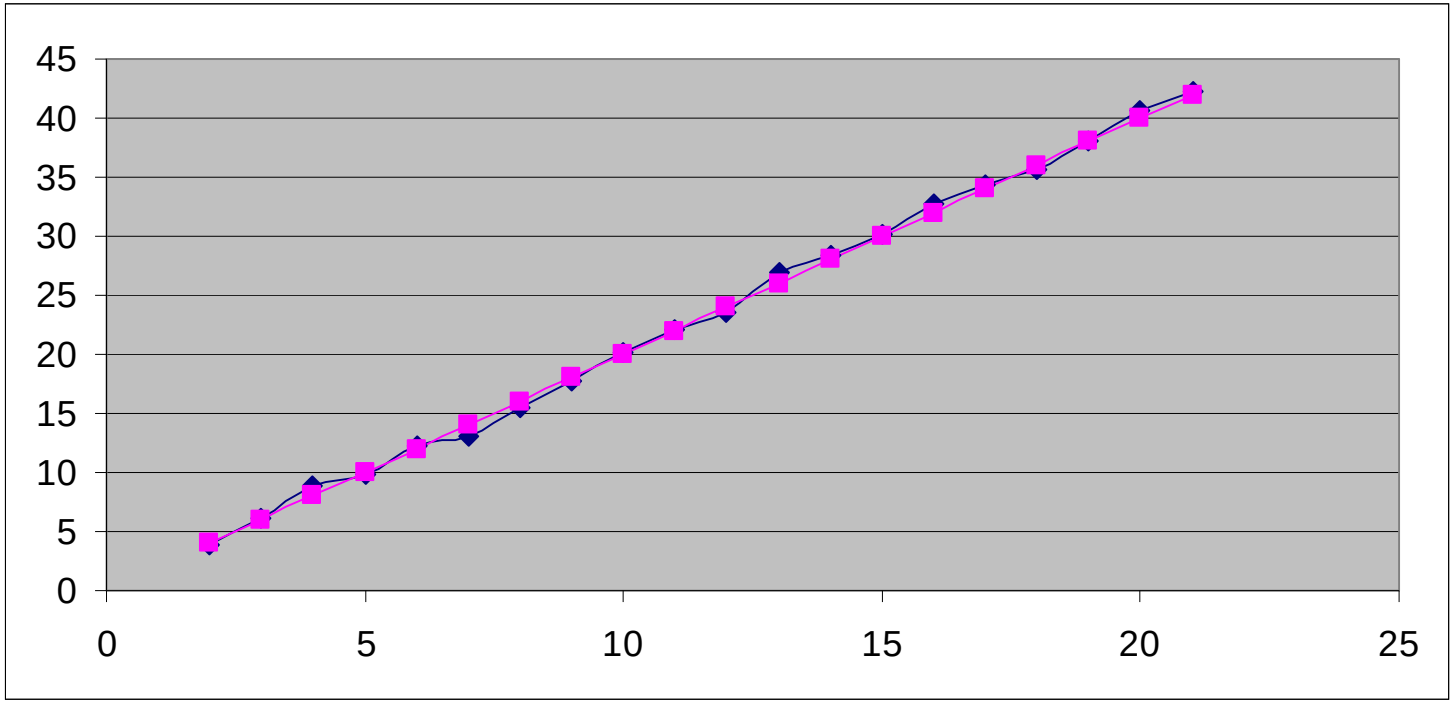
MSE = 169.7



Fit with m=1.5, and c = 0

$$y = m.x + c$$

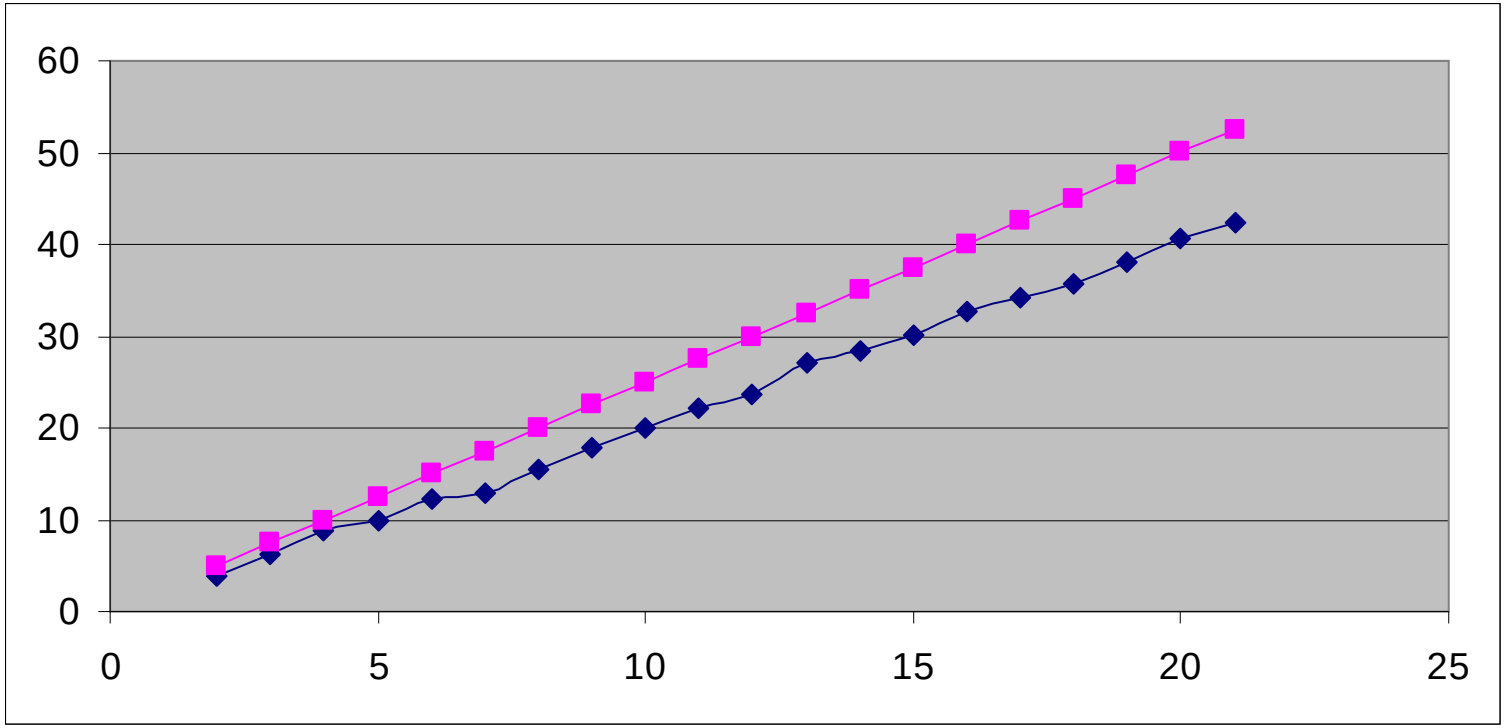
MSE = 43.6



Fit with m=2, and c = 0

$$y = m.x + c$$

MSE = 0.2365

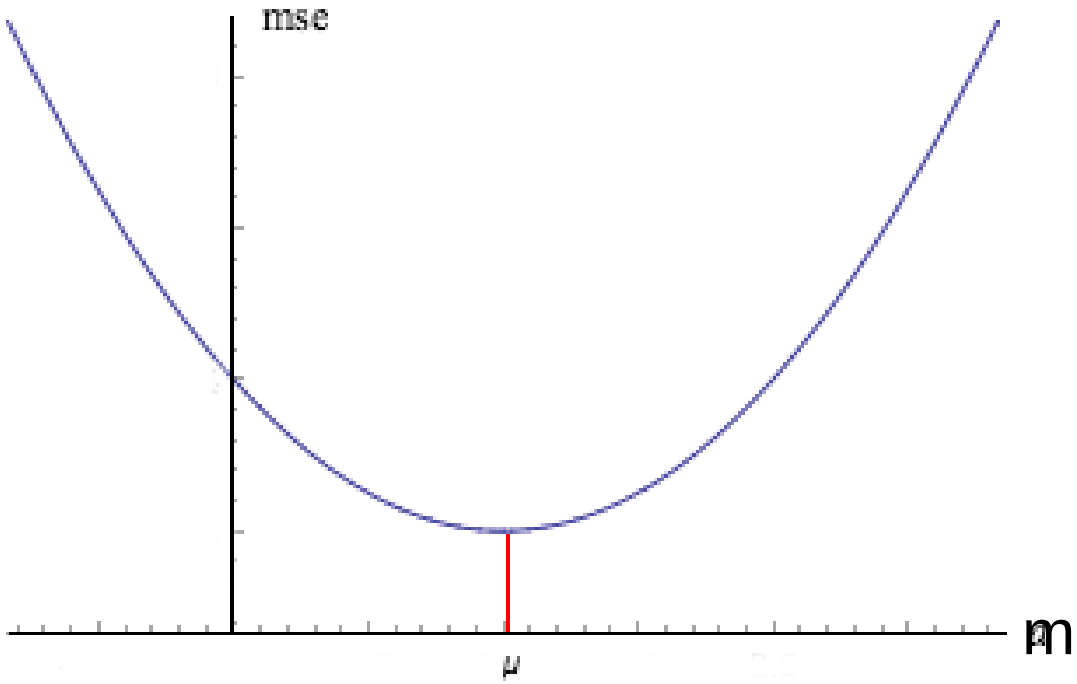


Fit with m=2.5, and c = 0

$$y = m.x + c$$

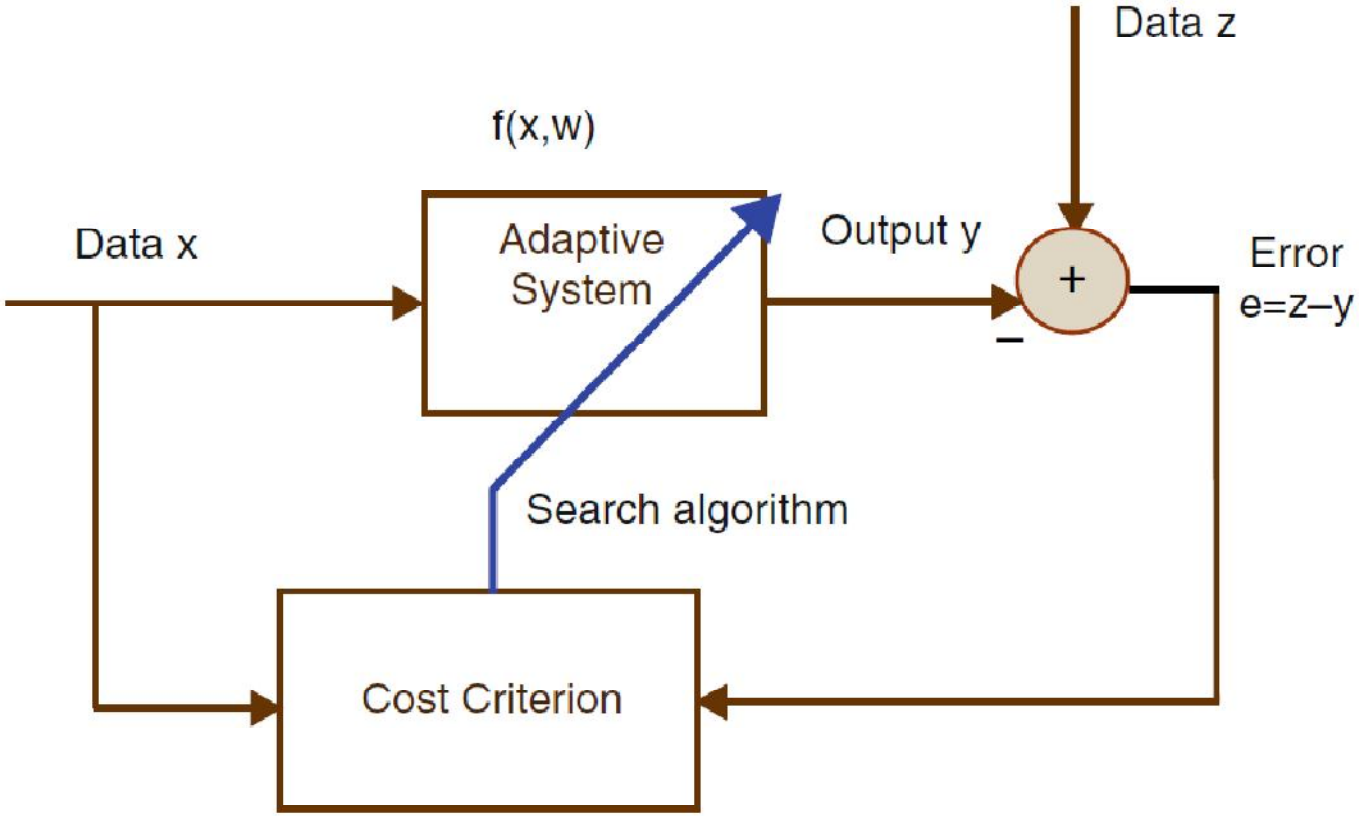
MSE = 39.6

Minimization of MSE



Adaptive system

- X: explanatory variable
- Y = f(x,w): generated output
- Z: response or desired output
- Error variable = z – f(x)

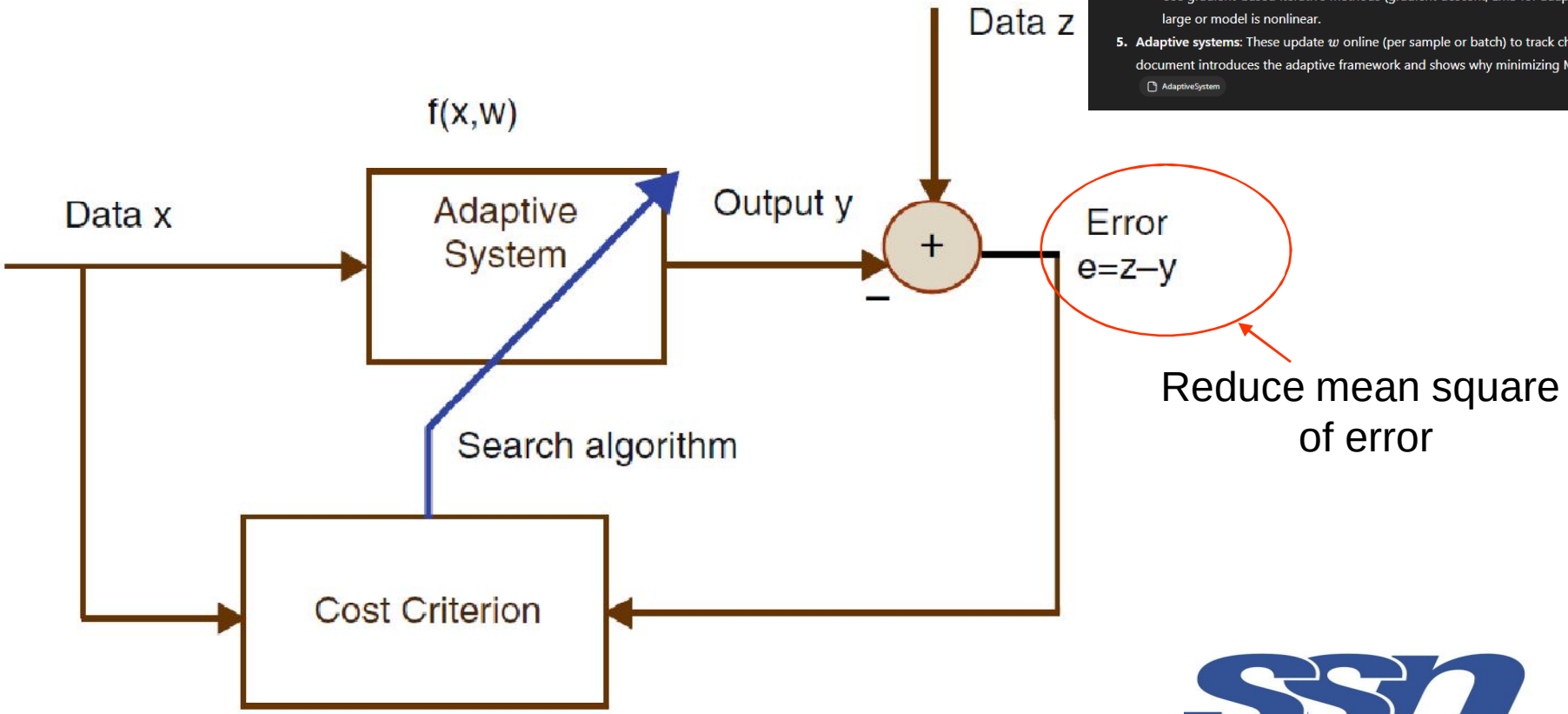


Error as a random variable

- X: explanatory variable
- Y = f(x,w): generated output
- Z: response or desired output
- Error variable = z – f(x)

Key mathematical points and practical notes (summary)

1. **Data & Model**: Given pairs (x_i, y_i), propose a model $f(x; w)$. Example: linear $f(x) = mx + c$. AdaptiveSystem
2. **Error**: For each point, error $e_i = y_i - f(x_i)$. We summarize with Sum of Squared Errors (SSE) or Mean Squared Error (MSE):

$$\mathrm{MSE}(w) = \frac{1}{K}\sum_{i=1}^{K}(y_i - f(x_i; w))^2.$$

3. **Cost surface**: MSE as a function of parameters w is often convex for linear-in-parameters models; it has a single global minimum that gives the best-fit parameters (in least-squares sense). Pages 6–9 illustrated how MSE changes as slope m changes. AdaptiveSystem
4. **Optimization**: To find optimal w, you can:
   - Solve analytically (normal equations) when possible (linear regression).
   - Use gradient-based iterative methods (gradient descent, LMS for adaptive filters) when data are large or model is nonlinear.
5. **Adaptive systems**: These update w online (per sample or batch) to track changing environments — the document introduces the adaptive framework and shows why minimizing MSE is the guiding principle. AdaptiveSystem